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Temperature control for insertable target holder for solid dopant materials

Abstract

An ion source with a target holder for holding a solid dopant material is disclosed. The ion source comprises a thermocouple disposed proximate the target holder to monitor the temperature of the solid dopant material. In certain embodiments, a controller uses this temperature information to vary one or more parameters of the ion source, such as arc voltage, cathode bias voltage, extracted beam current, or the position of the target holder within the arc chamber. Various embodiments showing the connections between the controller and the thermocouple are shown. Further, embodiments showing various placement of the thermocouple on the target holder are also presented.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/403,223 filed Aug. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/735,125 filed Jan. 6, 2020 (now U.S. Pat. No. 11,170,973 issued Nov. 9, 2021), which claims priority of U.S. Provisional Application Ser. No. 62/913,023 filed Oct. 9, 2019, the disclosures of which are incorporated herein by reference in their entireties.

FIELD
(1) Embodiments of the present disclosure relate to an ion source, and more particularly, an ion source with an insertable target holder to hold solid dopant materials, wherein the temperature of the dopant material or the target holder may be measured and optionally controlled.
BACKGROUND
(2) Various types of ion sources may be used to create the ions that are used in semiconductor processing equipment. For example, an indirectly heated cathode (IHC) ion source operates by supplying a current to a filament disposed behind a cathode. The filament emits thermionic electrons, which are accelerated toward and heat the cathode, in turn causing the cathode to emit electrons into the arc chamber of the ion source. The cathode is disposed at one end of an arc chamber. A repeller may be disposed on the end of the arc chamber opposite the cathode. The cathode and repeller may be biased so as to repel the electrons, directing them back toward the center of the arc chamber. In some embodiments, a magnetic field is used to further confine the electrons within the arc chamber. A plurality of sides is used to connect the two ends of the arc chamber.
(3) An extraction aperture is disposed along one of these sides, proximate the center of the arc chamber, through which the ions created in the arc chamber may be extracted.
(4) In certain embodiments, it may be desirable to utilize a material that is in solid form as a dopant species. For example, a crucible or target holder may be used to hold the metal such that when the metal liquefies, it remains in the target holder. Use of pure solid metal directly for ion implant increases the beam current available for wafer implant.
(5) However, there may be issues associated with the use of a target holder for solid dopant materials. For example, when using metals with low melting and boiling temperatures in a contained target holder, very high temperatures may be problematic. For example, the dopant material may become unstable and be prone to runaway effects that can cause inconsistent beam performance and lead to the undesired accumulation of dopant material in the arc chamber.
(6) Therefore, an ion source that may be used with solid dopant materials having low melting

temperatures, such as certain metals, and is capable of monitoring and controlling its internal temperature, would be beneficial.

SUMMARY

(7) An ion source with a target holder for holding a solid dopant material is disclosed. The ion source comprises a thermocouple disposed proximate the target holder to monitor the temperature of the solid dopant material. In certain embodiments, a controller uses this temperature information to vary one or more parameters of the ion source, such as arc voltage, cathode bias voltage, extracted beam current, or the position of the target holder within the arc chamber. Various embodiments showing the connections between the controller and the thermocouple are shown. Further, embodiments showing various placement of the thermocouple on the target holder are also presented.

(8) According to one embodiment, an indirectly heated cathode ion source is disclosed. The indirectly heated cathode ion source comprises an arc chamber, comprising a plurality of walls connecting a first end and a second end; an indirectly heated cathode disposed on the first end of the arc chamber; a target holder, having a pocket to hold a dopant material; a thermocouple in contact with the target holder; and a controller in communication with the thermocouple, wherein the controller varies a parameter of the ion source based on a temperature measured by the thermocouple. In certain embodiments, the ion source further comprises an actuator in communication with the target holder to move the target holder between a first position and a second position, and wherein the parameter comprises the position of the target holder. In some embodiments, the parameter is selected from the group consisting of arc voltage, filament current, cathode bias voltage, flow rate of feed gas, and beam extraction current. In certain embodiments, the ion source further comprises a heating element in communication with the target holder, and therein the parameter comprises the current supplied to the heating element. In some embodiments, the ion source further comprises an actuator assembly, the actuator assembly comprising: wires to electrically connect the thermocouple to the controller; a housing, comprising a rear housing, a front housing and an outer housing connecting the rear housing and the front housing; a shaft affixed to the target holder, and having a retaining plate disposed within the housing; a bellows disposed within the housing and affixed to the retaining plate on one end and to the rear housing on an opposite end; and an actuator to linearly translate the shaft. In some embodiments, a connector is mounted in the front housing, and the wires pass from the controller through a space between the outer housing and the bellows, and terminate at the connector. In some further embodiments, a second connector is mated to the connector, and thermocouple wires are disposed between the second connector and the thermocouple. In some embodiments, the thermocouple wires are coiled to allow target holder position adjustment with respect to the arc chamber. In some embodiments, the thermocouple wires are encased in an Inconel braid. In some embodiments, the thermocouple wires are encased in alumina tubes. In certain embodiments, the wires pass through a hollow interior of the shaft. In certain embodiments, the target holder comprises a target base, a crucible plug, a crucible and a porous plug. In some embodiments, the thermocouple is disposed on an outer surface of the crucible. In certain embodiments, a cavity is disposed on an interior surface of the target holder, and the thermocouple is disposed in the cavity on an outer surface of the crucible plug. In some embodiments, potting material is used to hold the thermocouple in place. In some embodiments, a set screw is used to hold the thermocouple in place. In certain embodiments, a spring is disposed in the cavity to hold the thermocouple in place. In some embodiments, the ion source comprises a heating element in communication with the target holder, wherein the heating element comprises resistive wires. In certain embodiments, the resistive wires are in communication with the crucible or the crucible plug.

(9) According to another embodiment, an assembly for use with an ion source is disclosed. The assembly comprises a connector; a thermocouple; and wires disposed between the connector and the thermocouple. In certain embodiments, the wires are coiled. In certain embodiments, the wires

are individually insulated. In some embodiments, the insulated wires are encased in an Inconel braid. In certain embodiments, the insulated wires are encased in alumina tubes.
(10) According to another embodiment, a target holder to hold a dopant material for use in an ion source is disclosed. The target holder comprises a target base; a crucible shaped as a hollow cylinder; a crucible plug to cover one open end of the crucible and disposed proximate the target base; a porous plug to cover an opposite end of the crucible, wherein gaseous dopant material may pass through the porous plug; and a thermocouple in communication with the target holder. In certain embodiments, the thermocouple is disposed on the outer surface of the crucible. In other embodiments, the thermocouple is disposed in a channel in the wall of the crucible. In some embodiments, there is a cavity disposed in the target base proximate the crucible plug and the thermocouple is disposed in the cavity proximate the crucible plug. In certain embodiments, a channel is disposed in the target base, wherein the channel in communication with the cavity to allow wires to be routed to the thermocouple. In some embodiments, potting material is used to hold the thermocouple in place. In certain embodiments, a set screw is used to hold the thermocouple in place. In some embodiments, the target holder comprises a shaft affixed to the target base. In some embodiment, an interior of the shaft is hollow to allow wires to be routed through the hollow interior of the shaft to the thermocouple.
(11) According to another embodiment, an indirectly heated cathode ion source is disclosed. The ion source comprises an arc chamber, comprising a plurality of walls connecting a first end and a second end; an indirectly heated cathode disposed on the first end of the arc chamber; a target holder, having a pocket to hold a dopant material, wherein the target holder is movable within the arc chamber; and a controller, wherein the controller varies a position of the target holder within the arc chamber based on a temperature of the dopant material. In some embodiments, the temperature of the dopant material is determined using optical measurements, a pyrometer, color dots, a thermocouple, a wireless thermocouple reader or an RTD (resistance temperature detector). In certain embodiments, the temperature of the dopant material is determined using a thermocouple. In some embodiments, the temperature of the dopant material is estimated based on a temperature of a component within the arc chamber.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) For a better understanding of the present disclosure, reference is made to the accompanying drawings, which are incorporated herein by reference and in which:
(2) FIG. **1** is an indirectly heated cathode (IHC) ion source with an insertable target holder in accordance with one embodiment;
(3) FIG. **2** shows the actuator assembly and target holder according to one embodiment;
(4) FIG. **3** shows the actuator assembly and target holder according to a second embodiment;
(5) FIG. **4** shows the actuator assembly and target holder according to a third embodiment;
(6) FIG. **5** shows the placement of the thermocouple on the target holder according to one embodiment;
(7) FIG. **6** shows the placement of the thermocouple on the target holder according to a second embodiment;
(8) FIG. **7** shows the placement of the thermocouple on the target holder according to another embodiment;
(9) FIGS. **8**A-**8**B show the placement of the thermocouple on the target holder according to other embodiments;
(10) FIGS. **9**A-**9**C show the placement of the thermocouple on the target holder according to other embodiments;

(11) FIG. **10** shows the actuator assembly and target holder according to one embodiment where resistive wires are used to heat the target holder;

DETAILED DESCRIPTION

(12) As noted above, at very high temperatures, the solid dopant in an ion source may melt too quickly and create unwanted accumulation of dopant in the arc chamber. At low temperatures, the solid dopant may not melt at all.

(13) FIG. **1** shows an IHC ion source **10** with a target holder that overcomes these issues. The IHC ion source **10** includes an arc chamber **100**, comprising two opposite ends, and walls **101** connecting to these ends. The walls **101** of the arc chamber **100** may be constructed of an electrically conductive material and may be in electrical communication with one another. In some embodiments, a liner may be disposed proximate one or more of the walls **101**. A cathode **110** is disposed in the arc chamber **100** at a first end **104** of the arc chamber **100**. A filament **160** is disposed behind the cathode **110**. The filament **160** is in communication with a filament power supply **165**. The filament power supply **165** is configured to pass a current through the filament **160**, such that the filament **160** emits thermionic electrons. Cathode bias power supply **115** biases filament **160** negatively relative to the cathode **110**, so these thermionic electrons are accelerated from the filament **160** toward the cathode **110** and heat the cathode **110** when they strike the back surface of cathode **110**. The cathode bias power supply **115** may bias the filament **160** so that it has a voltage that is between, for example, 200V to 1500V more negative than the voltage of the cathode **110**. The voltage difference between the cathode **110** and the filament **160** may be referred to as cathode bias voltage. The cathode **110** then emits thermionic electrons on its front surface into arc chamber **100**.

(14) Thus, the filament power supply **165** supplies a current to the filament **160**. The cathode bias power supply **115** biases the filament **160** so that it is more negative than the cathode **110**, so that electrons are attracted toward the cathode **110** from the filament **160**. In certain embodiments, the cathode **110** may be biased relative to the arc chamber **100**, such as by bias power supply **111**. The voltage difference between the arc chamber **100** and the cathode **110** may be referred to as arc voltage. In other embodiments, the cathode **110** may be electrically connected to the arc chamber **100**, so as to be at the same voltage as the walls **101** of the arc chamber **100**. In these embodiments, bias power supply **111** may not be employed and the cathode **110** may be electrically connected to the walls **101** of the arc chamber **100**. In certain embodiments, the arc chamber **100** is connected to electrical ground.

(15) On the second end **105**, which is opposite the first end **104**, a repeller **120** may be disposed. The repeller **120** may be biased relative to the arc chamber **100** by means of a repeller bias power supply **123**. In other embodiments, the repeller **120** may be electrically connected to the arc chamber **100**, so as to be at the same voltage as the walls **101** of the arc chamber **100**. In these embodiments, repeller bias power supply **123** may not be employed and the repeller **120** may be electrically connected to the walls **101** of the arc chamber **100**. In still other embodiments, a repeller **120** is not employed.

(16) The cathode **110** and the repeller **120** are each made of an electrically conductive material, such as a metal or graphite.

(17) In certain embodiments, a magnetic field is generated in the arc chamber **100**. This magnetic field is intended to confine the electrons along one direction. The magnetic field typically runs parallel to the walls **101** from the first end **104** to the second end **105**. For example, electrons may be confined in a column that is parallel to the direction from the cathode **110** to the repeller **120** (i.e. the y direction). Thus, electrons do not experience any electromagnetic force to move in the y direction. However, movement of the electrons in other directions may experience an electromagnetic force.

(18) Disposed on one side of the arc chamber **100**, referred to as the extraction plate **103**, may be an extraction aperture **140**. In FIG. **1**, the extraction aperture **140** is disposed on a side that is

parallel to the Y-Z plane (perpendicular to the page). Further, the IHC ion source **10** also comprises a gas inlet **106** through which the gas to be ionized may be introduced to the arc chamber **100**.
(19) In certain embodiments, a first electrode and a second electrode may be disposed on respective opposite walls **101** of the arc chamber **100**, such that the first electrode and the second electrode are within the arc chamber **100** on walls adjacent to the extraction plate **103**. The first electrode and the second electrode may each be biased by a respective power supply. In certain embodiments, the first electrode and the second electrode may be in communication with a common power supply. However, in other embodiments, to allow maximum flexibility and ability to tune the output of the IHC ion source **10**, the first electrode may be in communication with a first electrode power supply and the second electrode may be in communication with a second electrode power supply.
(20) A controller **180** may be in communication with one or more of the power supplies such that the voltage or current supplied by these power supplies may be modified. The controller **180** may include a processing unit, such as a microcontroller, a personal computer, a special purpose controller, or another suitable processing unit. The controller **180** may also include a non-transitory storage element, such as a semiconductor memory, a magnetic memory, or another suitable memory. This non-transitory storage element may contain instructions and other data that allows the controller **180** to perform the functions described herein.
(21) The IHC ion source **10** also includes a target holder **190**, which can be inserted into and retracted from the arc chamber **100**. In the embodiment of FIG. **1**, the target holder **190** enters the arc chamber along one of the walls **101** of the arc chamber **100**. In certain embodiments, the target holder **190** may enter the arc chamber **100** at the midplane between the first end **104** and the second end **105**. In another embodiment, the target holder **190** may enter the arc chamber **100** at a location different from the midplane. In the embodiment shown in FIG. **1**, the target holder **190** enters the arc chamber **100** through the side opposite the extraction aperture **140**. However, in other embodiments, the target holder **190** may enter through the sides that are adjacent to the extraction plate **103**. The target holder **190** may move between a first position and a second position.
(22) The target holder **190** has a cavity or pocket **191** into which the dopant material **195** may be disposed. The pocket **191** may have a bottom surface and sidewalls extending upward from the bottom surface. In certain embodiments, the sidewalls may be vertical. In other embodiments, the sidewalls may be slanted outward from the bottom surface. In some embodiments, the sidewalls and the bottom surface meet at a rounded edge. The bottom surface and the sidewalls form a cavity which is closed at the bottom. In other words, much like a traditional cup, the dopant material **195** is inserted or removed via the open top, while the sidewalls and bottom surface form a sealed structure from which the dopant material **195** cannot exit. In another embodiment, the pocket **191** may be enclosed, such that the dopant material **195** is disposed inside the pocket **191**. For example, a hollow cylindrical crucible may be employed to create the pocket **191**. A porous plug **192** may be used to hold the dopant material inside the pocket **191** and to allow vapors to exit the pocket **191**. The porous plug **192** may be graphite foam, for example. The feed rate of the dopant material from the target holder **190** may also be controlled by adding patterned holes of various size to the porous plug **192** or any other wall of the target holder **190**. Any of the walls of the target holder **190** may be a porous material and used for controlled feed of the dopant material into the arc chamber **100**.
(23) A dopant material **195**, such as indium, aluminum, antimony or gallium, may be disposed within the pocket **191** of the target holder **190**. The dopant material **195** may be in the form of a solid when placed in the pocket **191**. This may be in the form of a block of material, filings, shavings, balls, or other shapes. In certain embodiments, the dopant material **195** may melt and become a liquid. Therefore, in certain embodiments, the target holder **190** is configured to enter the arc chamber **100** such that the open end is facing upward and the sealed bottom is facing downward so that melted dopant material **195** cannot flow from the target holder **190** into the arc chamber **100**, but rather remains in the target holder **190**. In other words, the IHC ion source **10** and the target holder **190** are oriented such that the dopant material **195** is retained within the pocket **191**

by gravity.

(24) A thermocouple **198** may be in proximity to the target holder **190** or the dopant material **195**. This thermocouple **198** may be in communication with the controller **180**. The thermocouple **198** may comprise one or more wires **199** that electrically connect the thermocouple **198** to the controller **180**.

(25) In certain embodiments, the thermocouple **198** may be fixed to the outside of the target holder **190**. In other embodiments, the thermocouple **198** may include a rigid sheath that may be used to position relative to the target holder. In another embodiment, the thermocouple point of measurement may be directly inside the pocket **191**, holding the dopant material **195**. In these embodiments, the corrosion of thermocouple **198** may be prevented by using a ceramic insulator sheath to the protect the thermocouple wires.

(26) During operation, the filament power supply **165** passes a current through the filament **160**, which causes the filament **160** to emit thermionic electrons. These electrons strike the back surface of the cathode **110**, which may be more positive than the filament **160**, causing the cathode **110** to heat, which in turn causes the cathode **110** to emit electrons into the arc chamber **100**. These electrons collide with the molecules of gas that are fed into the arc chamber **100** through the gas inlet **106**. A carrier gas, such as argon, or an etching gas, such as fluorine, may be introduced into the arc chamber **100** through a suitably located gas inlet **106**. The combination of electrons from the cathode **110**, the gas and the positive potential creates a plasma. In certain embodiments, the electrons and positive ions may be somewhat confined by a magnetic field. In certain embodiments, the plasma is confined near the center of the arc chamber **100**, proximate the extraction aperture **140**. Chemical etching or sputtering by the plasma transforms the dopant material **195** into the gas phase and causes ionization. The ionized feed material can then be extracted through the extraction aperture **140** and used to prepare an ion beam.

(27) Negative ions and neutral atoms that are sputtered or otherwise released from the dopant material **195** are attracted toward the plasma, since the plasma is maintained at a more positive voltage than the target holder **190**.

(28) In certain embodiments, the dopant material **195** is heated and vaporized due to the heat created by the plasma. However, in other embodiments, the dopant material **195** may be heated by additional means as well. For example, a heating element **170** may be disposed within or on the target holder **190** to further heat the dopant material **195**. The heating element **170** may be a resistive heating element, or some other type of heater.

(29) In certain embodiments, the target holder **190** may be made of a conductive material and may be grounded. In a different embodiment, the target holder **190** may be made of a conductive material and may be electrically floated. In a different embodiment, the target holder **190** may be made of a conductive material and may be maintained at the same voltage as the walls **101**. In other embodiments, the target holder **190** may be made of an insulating material.

(30) In yet another embodiment, the target holder **190** may be biased electrically with respect to the arc chamber **100**. For example, the target holder **190** may be made from a conductive material and may be biased by an independent power supply (not shown) so as to be at a different voltage than the walls **101**. This voltage may be more positive or more negative than the voltage applied to the walls **101**. In this way, electrical biasing may be used to sputter the dopant material **195**.

(31) The controller **180** may monitor the temperature of the dopant material **195** using the thermocouple **198**. In certain embodiments, the controller **180** may be in communication with a thermocouple **198** and with the heating element **170**. Thus, the controller **180** may control the heating element **170** to maintain the dopant material **195** at a desired or predetermined temperature. In other words, the controller **180** may vary the current through the heating element **170** to maintain a desired temperature, as measured by the thermocouple **198**. This may allow the controller **180** to control the feed rate of dopant material **195** into the arc chamber **100**. In other embodiments, the controller **180** may indirectly measure the temperature of the dopant material

195, such as by measuring the temperature of the target holder **190** or some other component.

(32) The target holder **190** is in communication with one end of shaft **200**. The opposite end of the shaft **200** may be in communication with an actuator assembly **300**. The actuator assembly **300** may be attached directly to one of the walls **101**. In other embodiments, the actuator assembly **300** may be set back from the wall **101** to allow the target holder **190** to be retracted out of the main cylinder of the arc chamber **100**. The actions of the actuator assembly **300** allow the target holder **190** to move linearly within the arc chamber **100**.

(33) FIG. **2** shows one embodiment of the actuator assembly **300**. In this embodiment, the actuator assembly **300** includes a rear housing **310** and a front housing **340**. The front housing **340** may be bolted or otherwise connected to one of the walls **101** of the arc chamber **100**. Alternatively, the front housing **340** may be set back from the walls **101**. An outer housing **360** may be used to connect the rear housing **310** and the front housing **340**.

(34) Inside the rear housing **310** is an actuator **320**. The actuator **320** may have a drive shaft **325**. In certain embodiments, the actuator **320** is an electric motor, although other types of actuator may be used. In one embodiment, the drive shaft **325** has a threaded distal end **326**. A correspondingly threaded member **330** may be in communication with the threaded distal end **326**. The threaded member **330** may be affixed to the shaft **200**. In this way, when the drive shaft **325** rotates, the threaded member **330** is drawn to the actuator **320** or moved away from the actuator **320**, depending on the direction of rotation. Since the shaft **200** is affixed to the threaded member **330**, the shaft **200** similarly is translated linearly in the X direction by the rotational movement of the drive shaft **325**. This allows the target holder **190** to be disposed in different locations within the arc chamber **100**.

(35) In this embodiment, the shaft **200** includes a retaining plate **210**. The retaining plate **210** is disposed inside the actuator assembly **300** behind the front housing **340**. The retaining plate **210** is welded or otherwise connected to bellows **350**. In certain embodiments, the bellows **350** may be metal. The bellows **350** may also be welded or otherwise attached to the rear housing **310**. The bellows **350** and the retaining plate **210** form the barrier between the vacuum conditions in the arc chamber **100** and the atmospheric conditions outside of the arc chamber **100**. Thus, when the drive shaft **325** rotates, the bellows **350** expands and contracts based on the direction of motion of the shaft **200**.

(36) Note that the thermocouple **198** is disposed inside the arc chamber **100**, and the wires **199** need to exit the arc chamber **100**, while preserving the integrity of the vacuum conditions. In the embodiment of FIG. **2**, a first connector **390** is mounted inside the arc chamber **100** on the front housing **340**. Wires **199** extend from outside the actuator assembly **300** to the first connector **390**. In this embodiment, a channel **311** may be created in the rear housing **310** to allow the wires **199** to pass out of the actuator assembly **300**. The wires **199** may then pass in the space between the bellows **350** and the outer housing **360**. This space is part of the vacuum environment, and therefore a vacuum feedthrough **370** is used to maintain the vacuum. A vacuum feedthrough is a member that allows the passage of wires **199**, but maintains the pressure difference between the two sides of the feedthrough. Thus, the wires **199** pass through channel **311** in the rear housing **310**, then pass through the vacuum feedthrough **370**. The wires **199** then pass through the space between the outer housing **360** and the bellows **350** and finally terminate at the first connector **390**.

(37) A second connector **391** mates with the first connector **390**. Thermocouple wires **197** extend from the second connector **391** to the thermocouple **198**. The thermocouple **198** may be a K type thermocouple. Further, the thermocouple wires **197** attached to the thermocouple **198** may be insulated. For example, in one embodiment, each of the two thermocouple wires **197** is individually coated with an insulating material. The two thermocouple wires **197** may then be wrapped together in an Inconel braid. In other words, the thermocouple wires **197** are individually coated for electrical insulation and the pair are then wrapped to protect them from the harsh environment in the arc chamber **100**. In another embodiment, the thermocouple wires **197** may be encased in

alumina tubes.
(38) In certain embodiments, the thermocouple wires **197** are coiled, as shown in FIG. **2**. In this way, when the target holder **190** is extended and retracted, the thermocouple wires **197** coil and uncoil to compensate for the change in length.
(39) In one embodiment, the thermocouple **198**, the thermocouple wires **197** and the second connector **391** may be a replaceable part. Further, as described above, the thermocouple wires **197** is this embodiment may be individually insulated and then wrapped in a braid. Further, the thermocouple wires **197** may be coiled to allow for changes in length without kinking or interference.
(40) A heating element **170** may be disposed within or on the target holder **190** to further heat the dopant material **195**. In certain embodiments, wires associated with the heating element **170** are routed with the thermocouple wires **197**.
(41) The thermocouple **198** may be attached to the target holder **190** in a plurality of ways, which are described below.
(42) FIG. **3** shows a second embodiment of an actuator assembly **300**. Many of the components are the same as shown in FIG. **2** and have been given identical reference designators. In this embodiment, the shaft **200** may be hollow such that thermocouple wires **197** may be routed through an interior of the shaft **200**. The shaft **200** also has an opening **201** to the hollow interior. The opening **201** may be located on the side of the retaining plate **210** further from the target holder **190**. In this way, the opening **201** is in atmospheric conditions. If the thermocouple **198** is located within the hollow interior of the shaft **200**, a vacuum feedthrough may not be needed. However, if the thermocouple **198** is located on an exterior surface of the target holder **190**, such as shown in FIG. **4**, a vacuum feedthrough **370** may be used to preserve the vacuum within the arc chamber **100**. The vacuum feedthrough **370** would be disposed at the entrance to the hollow interior of the shaft **200**.
(43) The thermocouple wires **197** pass through the opening **201** and may exit through the channel **311** in the rear housing **310**. In certain embodiments, one or more cable mounts **351** may be used to hold the thermocouple wires **197** in place. In certain embodiments, the wires **199** that are in communication with the controller **180** are the same as the thermocouple wires **197** that pass through the hollow interior of the shaft **200**. In other embodiments, a connector may be disposed between the thermocouple **198** and the controller **180** to create two separate wire segments. For example, the portion of the thermocouple wires that are exposed to the plasma may need to be replaced more often. Therefore, this section of the wires may be formed as a replaceable unit by inserting a connector between the thermocouple **198** and the controller **180**.
(44) Thus, in this embodiment, the shaft has a hollow interior that is used to route the thermocouple wires **197** from the thermocouple **198** to the interior of the actuator assembly **300**. As stated, a vacuum feedthrough **370** may be employed at the entrance to the interior of the shaft **200** if the thermocouple **198** is disposed in vacuum conditions, as shown in FIG. **4**.
(45) In certain embodiments, the thermocouple wires **197** are individually insulated and then wrapped together in an Inconel braid or alumina tubes. In other embodiments, because the thermocouple wires are protected by the shaft **200**, an Inconel braid is not employed.
(46) FIGS. **2-4** describe several systems that may be used to route the wires from the controller **180** to the thermocouple **198**. FIGS. **5-9** show various embodiments concerning the placement of the thermocouple **198** on the target holder **190**.
(47) FIG. **5** shows an enlarged view of the target holder **190**. In certain embodiments, the target holder **190** comprises a target base **193**, which is affixed or otherwise attached to the shaft **200**. The target holder **190** may also include a crucible **196**. The crucible **196** holds the dopant material **195**. In certain embodiments, the crucible **196** may be made from graphite. In some embodiments, the crucible **196** may be a hollow cylinder having two open ends. A crucible plug **194** may be disposed between the crucible **196** and the target base **193**. The crucible plug **194** is used to plug one of the

open ends of the crucible **196**. Clamp **410** may be used to secure the target base **193** to the crucible **196**. As described above, a porous plug **192** may be used to plug the second open end of the crucible **196**. As stated above, this porous plug **192** may be made from graphite foam or another suitable material.

(48) In the embodiment of FIG. **5**, the thermocouple **198** is mounted on an outer surface of the crucible **196**. Potting material **400** may be used to hold the thermocouple **198** in place. The thermocouple wires **197** may be routed along the exterior of the target holder **190**.

(49) FIG. **6** shows another embodiment of the target holder **190**. Many of the components are the same as shown in FIG. **5** and have been given identical reference designators. In this embodiment, a conduit **420** is created in the target base **193** and optionally in the crucible plug **194**. The thermocouple wires **197** pass through the conduit **420** and the thermocouple **198** is mounted on the outer surface of the crucible **196**, as was done in FIG. **5**. Potting material **400** may be used to hold the thermocouple **198** in place.

(50) FIG. **7** shows another embodiment of the target holder **190**. Many of the components are the same as shown in FIG. **6** and have been given identical reference designators. In this embodiment, rather than using potting material, a set screw **430** is used to hold the thermocouple **198** in place. The set screw **430** may screw into a threaded shallow hole in the crucible **196**. In some embodiments, the threaded shallow hole does not pass through to the interior of the crucible **196**.

(51) It is noted that the set screw **430** may be used with the embodiment of FIG. **5**. In other words, the thermocouple wires **197** may be routed around the exterior of the target holder **190** and be secured to the crucible **196** using set screw **430**.

(52) In summary, FIGS. **5-7** show different target holders **190** where the thermocouple **198** is in contact with an outer surface of the crucible **196**. This thermocouple **198** may be affixed to the crucible **196** using a potting material **400** or a set screw **430**. Other fastening techniques may also be employed.

(53) Additionally, the thermocouple **198** may be embedded in the wall of the crucible **196**. FIG. **8**A shows an embodiment where a channel **440** is created in the wall of the crucible. Many of the components are the same as shown in FIG. **6** and have been given identical reference designators. The channel **440** is narrower than the width of the wall of the crucible **196**. The thermocouple **198** is inserted into the channel **440**. Potting material (not shown) may be used to hold the thermocouple **198** in place. In this embodiment, the channel **440** may extend through the target base **193** and optionally the crucible plug **194**. In another embodiment, shown in FIG. **8**B, the channel **440** exits on the outer surface of the crucible **196**. In this embodiment, the channel does not extend through the target base **193** or the crucible plug **194**.

(54) In another embodiment, the channel **440** may extend to the pocket **191**, so that the thermocouple **198** is actually in contact with an interior of the pocket **191** and/or the dopant material **195**. In these embodiments, a ceramic insulator sheath may be employed to the protect the thermocouple **198** and the thermocouple wires **197**.

(55) The thermocouple **198** may also be in contact with the crucible plug **194**, as shown in FIGS. **9**A-**9**C. Many of the components are the same as shown in FIG. **6** and have been given identical reference designators. In these embodiments, a cavity **450** is disposed in the target base **193**. The cavity **450** provides a location where the thermocouple **198** can be placed. A channel **460** is created in the target base **193** from an exterior of the target base **193** to the cavity **450**. Thermocouple wires **197** enter the cavity **450** via the channel **460**. FIG. **9**A shows the thermocouple **198** held in place through the use of potting material **400**. FIG. **9**B shows the thermocouple **198** held in place using a set screw **430**. FIG. **9**C shows the thermocouple **198** held in place through the use of a spring **470**. Of course, other force-based means may also be used to hold the thermocouple **198** in place.

(56) While the above disclosure describes various apparatus for routing the wires for the thermocouple **198** to the target holder **190**, the same techniques may also be used to route resistive wires to the target holder **190**. These resistive wires may be employed as a heating element **170**.

For example, resistive wires may be in contact with all or a portion of the outer surface of the crucible **196**, as shown in FIGS. **1-4**. The resistive wires may be routed using the same means as shown in FIGS. **2-7**. Alternatively, the resistive wires may be embedded in the wall of the crucible, similar to the embodiments shown in FIGS. **8**A-**8**B. In another embodiment, the resistive wires may be in contact with the crucible plug **194**, such as shown in FIGS. **9**A-**9**C. When a current is passed through the resistive wires, heat is generated. This may allow the controller **180** another mechanism to control the temperature of the dopant material **195**.
(57) In certain embodiments, the resistive wires are bundled with the thermocouple wires **197**. In these embodiments, the resistive wires are routed with the thermocouple wires **197**.
(58) In other embodiments, the resistive wires are provided in a separate braid or bundle, and traverse the same path as the thermocouple wires **197**.
(59) In yet other embodiments, the resistive wires may be in contact with one part of the target holder **190**, such as the crucible **196**, while the thermocouple **198** is in contact with another part of the target holder **190**, such as the crucible plug **194**. FIG. **10** shows an embodiment where the resistive wires **500** are separate from the thermocouple wires **197**. This embodiment is similar to FIG. **2**, but includes a third connector **510** and a fourth connector **520**. The third connector **510** may be mounted to the front housing **340**. Wires **540** from the controller **180** are routed to the third connector **510**. In certain embodiments, the components used for routing the wires **540** are similar to those for routing the wires **199**. For example, a channel **311** and vacuum feedthrough **370** may be employed. The resistive wires **500** may be coiled to allow changes in length and may be connected to an outer surface of the crucible or to the crucible plug.
(60) Although FIG. **10** shows a third and fourth connector, it is understood that the resistive wires **500** may also be routed with the wires **199** and a larger connector may be used.
(61) Alternatively, the resistive wires **500** may also be routed through the shaft **200**, similar to the routing of wires **199** shown in FIGS. **3-4**.
(62) The above disclosure describes various embodiments that allow the controller **180** to monitor the temperature of the dopant material **195** by measuring the temperature of a component (i.e. the crucible, the crucible plug, etc.) using a thermocouple **198**. The controller **180** may use this information in a variety of ways.
(63) It may be advantageous to heat the dopant material **195** to a temperature within a predetermined range. For example, at low temperatures, the dopant material **195** may not melt, and therefore no dopant vapor is released from the target holder **190**. However, at excessively high temperatures, the melt rate of the dopant material may be too great. This may cause accumulation of dopant material in the arc chamber **100**. Additionally, variation in the melt rate may also affect the beam current and other parameters.
(64) By monitoring the temperature at or near the target holder **190**, the controller **180** may be able to better regulate the temperature of the dopant material **195**. For example, the controller **180** may monitor the temperature of the dopant material **195**. If the temperature is not within the predetermined range, the controller may change the current through filament power supply **165**, change the arc voltage, change cathode bias voltage, alter the flow rate of gas into the arc chamber **100**, change the position of the target holder **190** in the arc chamber **100**, vary the beam extraction current or perform a combination of these actions. Furthermore, if first and second electrodes are disposed on walls **101**, the voltage applied to these electrodes may also be varied by the controller **180** based on the temperature of the dopant material **195**. Additionally, in embodiments where a heater is employed, such as through the use of resistive wires **500**, the controller **180** may vary the current through the heater to change the temperature of the dopant material **195**.
(65) In certain embodiments, the controller **180** may move the position of the target holder **190** within the arc chamber **100** based on the temperature of the dopant material **195**. For example, the target holder **190** may heat to a higher temperature when disposed directly in the cylindrical region defined between the cathode **110** and the repeller **120**. To slow the heating of the dopant material,

the target holder **190** may be moved linearly to be outside this cylindrical region. Conversely, to increase the temperature of the dopant material **195**, the target holder **190** may be moved into this cylindrical region.

(66) The controller **180** may employ various closed loop algorithms to determine the parameters associated with the IHC ion source **10** based on the temperature obtained by the thermocouple **198**.

(67) While the above disclosure describes the use of a thermocouple **198**, other temperature sensors may also be used. For example, optical measurements, a pyrometer, and color dots are all indirect methods of detecting the temperature of the target holder **190**. RTDs (resistance temperature detectors) and wireless thermocouple readers may also be employed. Thus, the above disclosure is not limited to the use of thermocouples.

(68) Further, while the above disclosure describes the thermocouple **198** as being in contact with the target holder **190** or the dopant material **195**, other embodiments are also possible. For example, the thermocouple **198** (or other temperature sensor) may measure the temperature of another component within the arc chamber **100** and estimate the temperate the of the dopant material based on this measured temperature. This other component may be a wall of the arc chamber **100**, the shaft **200**, the repeller **120**, the front housing **340**, or another component.

(69) Further, the control described above may be performed using open loop techniques. For example, empirical data may be collected to determine the temperature of the dopant material as a function of various parameters, such as cathode bias voltage, arc voltage, feed gas flow rate, and position of the target holder. The empirical data may also determine the temperature of the dopant material as a function of time. Using tables or equations, the controller **180** may vary one or more of the parameters to maintain the dopant material **195** within a predetermined range. For example, the controller **180** may move the target holder **190** over time to maintain its temperature within the desired range.

(70) While the above disclosure describes the use of a target holder in an indirectly heated cathode ion source, the disclosure is not limited to this embodiment. The target holder, actuator assembly and thermocouple may also be employed in other ion or plasma sources, such as capacitively coupled plasma sources, inductively coupled plasma sources, a Bernas source or another suitable source.

(71) The embodiments described above in the present application may have many advantages. Using an insertable target holder allows for pure metal dopants to be used as sputter targets in an environment that exceeds their melting temperature. Traditionally, an oxide/ceramic or other solid compound containing the dopant that has a melting temperature of greater than 1200° C. is used. The use of a dopant-containing compound rather than a pure material severely dilutes the available dopant material. For example, when using Al.sub.2O.sub.3 as an alternative to pure aluminum, the stoichiometry of the ceramic composition not only introduces impurities into the plasma, possibly introducing undesirable mass coincidences with the dopant of interest, but also leads to lower beam currents than a pure elemental target. Use of Al.sub.2O.sub.3 may also lead to generation of undesirable byproducts, such as oxides and nitrides, which can be deposited along the beamline and compromise the operation of the ion implanter. For instance, the beam optics may be cleaned chemically to maintain ion beam stability after use of Al.sub.2O.sub.3.

(72) In one experiment, beam currents of up to 4.7 mA were achieved using a pure Al sputter target, whereas a maximum beam current of less than 2 mA could be achieved using an Al.sub.2O.sub.3 target. Use of pure metals will also increase the multi charge beam currents by 50%-75% as compared to the beam currents obtained from oxides/ceramics of the same metal species. With the insertable container, access to a large volume of pure metal is available when needed and the solid target can be safely removed from the arc chamber to utilize other species.

(73) Further, by monitoring the temperature of the dopant material, the ion source can be controlled to ensure that the melt rate of the dopant material is within a predetermined range. Specifically, without temperature control, it is possible that the dopant feed may become unstable and prone to

runaway effect that can cause inconsistent beam performance and lead to undesired accumulation of dopant material in the arc chamber. Thus, temperature control prevents exponential increase in dopant vapor in the arc chamber. It also allows faster tuning of the ion source.
(74) Additionally, by monitoring the temperature of the dopant material, it is possible to use this information in the beam tuning process, thus making beam performance more reliable.
(75) The present disclosure is not to be limited in scope by the specific embodiments described herein. Indeed, other various embodiments of and modifications to the present disclosure, in addition to those described herein, will be apparent to those of ordinary skill in the art from the foregoing description and accompanying drawings. Thus, such other embodiments and modifications are intended to fall within the scope of the present disclosure. Furthermore, although the present disclosure has been described herein in the context of a particular implementation in a particular environment for a particular purpose, those of ordinary skill in the art will recognize that its usefulness is not limited thereto and that the present disclosure may be beneficially implemented in any number of environments for any number of purposes. Accordingly, the claims set forth below should be construed in view of the full breadth and spirit of the present disclosure as described herein.

Claims

1. An indirectly heated cathode ion source, comprising: an arc chamber, comprising a plurality of walls connecting a first end and a second end; an indirectly heated cathode disposed on the first end of the arc chamber; a shaft extending into the arc chamber; and a target holder comprising: a target base affixed to the shaft; a crucible shaped as a hollow cylinder, wherein one end of the crucible is disposed proximate the target base; and a porous plug to cover an open end of the crucible, wherein gaseous dopant material may pass through the porous plug.
2. The indirectly heated cathode ion source of claim 1, further comprising a crucible plug to cover an opposite open end of the crucible and disposed proximate the target base.
3. The indirectly heated cathode ion source of claim 1, further comprising a thermocouple in communication with the target holder.
4. The indirectly heated cathode ion source of claim 3, wherein the thermocouple is affixed to an outer surface of the crucible.
5. The indirectly heated cathode ion source of claim 3, wherein the thermocouple is disposed in a channel in a wall of the crucible.
6. The indirectly heated cathode ion source of claim 3, further comprising a crucible plug to cover an opposite open end of the crucible and disposed proximate the target base, wherein a cavity is disposed in the target base proximate the crucible plug and the thermocouple is disposed in the cavity proximate the crucible plug.
7. The indirectly heated cathode ion source of claim 6, wherein a channel is disposed in the target base, the channel in communication with the cavity to allow wires to be routed to the thermocouple.
8. The indirectly heated cathode ion source of claim 3, wherein an interior of the shaft is hollow to allow wires to be routed through a hollow interior of the shaft to the thermocouple.
9. The indirectly heated cathode ion source of claim 1, further comprising a heating element in communication with the target holder, wherein the heating element comprises resistive wires.
10. The indirectly heated cathode ion source of claim 1, wherein the porous plug comprises graphite foam.
11. The indirectly heated cathode ion source of claim 1, further comprising a clamp to secure the crucible to the target base.

12. The indirectly heated cathode ion source of claim 1, further comprising an actuator in communication with the shaft to move the target holder within the arc chamber.